

Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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Abstract—We study online control of a scalar system under adversarial disturbances, convex stage costs, and state-dependent constraints. At each time, the controller must choose a feasible action before observing that time’s disturbance and cost. We introduce *Simplex Disturbance-Action Control* (SDAC), a simplex-constrained disturbance-action policy class that preserves per-step feasibility by construction. An online controller updates SDAC parameters via exponentiated gradient on the subprobability simplex, with a truncation step that enforces control feasibility. We prove regret that is sublinear in the horizon and logarithmic in the policy dimension against the best fixed SDAC policy in hindsight, implying no-regret relative to feasible linear state-feedback policies. The setting is motivated by energy-harvesting battery management and similar storage systems.

Index Terms—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state constraints.

I. INTRODUCTION

A. Motivation and Problem Setting

MANY dynamic resource systems can be modeled by a storage state that evolves over time and constrains the control action at every time. The controller must choose an action subject to state-dependent constraints before observing that time’s disturbance and cost.

This creates a constrained discrete-time control problem with state–control coupling: aggressive early control actions can harm later performance, while overly conservative actions can increase immediate cost. Our objective is to design a policy that remains feasible at every time and competes with the best fixed policy in a structurally tractable policy class.

B. Model and Information Structure

We study a discrete-time storage system with state $x_t \geq 0$, where x_t models the storage level at the beginning of time t . At each time t , the controller chooses a control action u_t under the *state constraints*

$$0 \leq u_t \leq \alpha x_t,$$

where $\alpha \in (0, 1)$ is a fixed known state-decay (retention) factor. After the action is chosen, an adversarial disturbance w_t arrives, and the state evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t.$$

Unlike stochastic application models that rely on distributional assumptions, our formulation is adversarial. Disturbances w_t and costs $c_t(x, u)$ may vary arbitrarily over time, and the controller chooses u_t before observing either at time t . The goal is low cumulative cost relative to the best fixed policy in a later-defined policy class, while both our algorithm and the benchmark policy maintain feasibility throughout.

C. Application Domains

This abstraction captures a broad set of domains:

- **Energy Harvesting and Battery Management:** x_t is battery state of charge (with self-discharge factor α), w_t is harvested energy, and u_t is discharge or consumption bounded by available energy αx_t .
- **Queueing Networks and Buffer Management:** x_t is queue/buffer occupancy (with abandonment/leakage factor α), w_t is packet arrivals, and u_t is service bounded by αx_t .
- **Inventory Control:** x_t is on-hand inventory (with spoilage/deterioration factor α), w_t is supply or replenishment, and u_t is sales bounded by αx_t .
- **Investments and Bank Accounts:** x_t is liquid capital/balance (with fee/retention factor α), w_t is incoming cash flow, and u_t is capital deployment or withdrawal bounded by αx_t .
- **Water Reservoir and Dam Management:** x_t is stored water (with evaporation/seepage factor α), w_t is inflow, and u_t is controlled release bounded by αx_t .

a) *Energy Harvesting and Battery Management:* A closely related work is [1], which studies an online energy-harvesting allocation problem with dynamics similar to ours. That work benchmarks against the best sequence of decisions in hindsight, but it differs in several important ways: [1] uses competitive ratio rather than regret, adopts an action-only cost model rather than general convex state-action costs, which means in particular that it cannot optimize a function of the battery size, and it has a less adversarial information pattern because the algorithm observes the current cost before selecting the action, whereas in our model the control action is chosen before the cost is revealed.

Prior work on energy harvesting includes [2], which improves over [3] by achieving $O(\sqrt{T})$ regret with finite battery and adversarial costs (versus the stochastic costs in [3]). Both works consider action-only costs (not state-dependent), i.i.d. exogenous disturbances (not adversarial), and benchmark against fixed allocation amounts rather than fixed policies. In contrast, our framework jointly optimizes over state-action

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stage costs, fully adversarial sequences, and stronger benchmarks, requiring new algorithmic techniques.

b) *Queueing and Network Optimization*: An alternative, though mathematically related, perspective comes from queueing control in computer networks and telecommunications. In that framework, one tracks a queue with stochastic arrivals and makes real-time service decisions, typically aiming to minimize long-run average service cost while keeping queue lengths stable. This literature is extensive (see, e.g., [4]–[7]), but its setting differs from ours in several important respects. Classical queueing work usually assumes stochastic arrivals and focuses on simple, known cost functions that depend only on the service rate, while queue length is kept small. These methods achieve trade-offs between queue size and time-average service cost. By contrast, our model permits fully adversarial sequences of exogenous disturbances and stage costs and allows general convex costs that depend on both the queue length (state) and the service rate (control action). Thus our approach is more general in cost structure and exogenous-disturbance model. The queueing literature provides useful motivation and techniques for state-coupled dynamics, but does not directly address the online optimization and regret-minimization objectives we study.

Another related work is [8], which proposes and analyzes a fixed controller that is a linear function of past arrivals. That controller resembles our family of SDAC policies because it uses nonnegative filter weights that sum to one (summing to at most one for SDAC). However, the controller in [8] is not updated online and is chosen to achieve bounded service delay; by contrast, our framework focuses on online optimization of a general cost and does not treat delay as a constraint.

c) *Inventory Management*: A related work is [9], which studies a distinct model in which demand (equivalent to allocation in our setup) is chosen adversarially, while the controller chooses the supply (equivalent to exogenous disturbances in our model). The objective is to minimize time-average purchasing cost while always meeting demand, and the performance metric is competitive ratio rather than regret. Although structurally similar to our setting, the control variable differs: we choose the control action from the feasible state bound, whereas [9] chooses how much to procure. Another key difference is the cost model: our framework allows general convex state-action stage costs, while [9] uses a linear cost depending only on the control action.

More closely related to ours is [10], which considers an inventory-control formulation with potentially nonlinear and nondifferentiable state-transition functions, and studies base-stock policies. While that model is more general in scope, its decision structure differs from ours: as in [9], the controller chooses supply (inflow) and then observes demand (allocation) through the transition, whereas our controller chooses the control action and observes incoming supply (exogenous disturbances). Methodologically, our setup is intentionally narrower and tailored to SDAC-based adversarial regret analysis: feasibility constraints are enforced by policy design, yielding a simpler linear state recursion, lower computational overhead, and cleaner analysis than approaches operating through nonlinear nondifferentiable dynamics such as [10]. The two

frameworks are structurally related but target different problem structures, control variables, and policy classes.

d) *Water Reservoir and Dam Management*: Classical dam theory studies storage dynamics with random inflows and controlled releases (see, for example, [11], [12]). Relative to our adversarial online-control setting, this literature typically assumes stochastic inputs and emphasizes stationary performance criteria under simple, known cost structures. Here, it serves mainly as historical context for state-coupled control under state constraints.

e) *Online control*: A recent line of work in online learning studies online control, where a linear dynamical system is operated under adversarial disturbances and costs. The objective is low time-average cost relative to the best policy in hindsight, rather than classical stabilization or worst-case guarantees. Foundational results for linear systems and adversarial settings include [13]–[16]; see [17] for a modern overview.

A central idea in this literature is *Disturbance-Action Control* (DAC), a policy parameterization in which control actions are expressed as affine or linear functions of the current state and a finite history of past disturbances. This representation yields tractable surrogates for policy optimization and enables efficient no-regret updates in adversarial environments. Extensions include partial observability [18] and bandit feedback [15]. SDAC adapts this template to state-constrained resource systems, where feasibility is coupled over time through the dynamics.

Online control provides a strong foundation for our problem class: it accommodates arbitrary time-varying disturbances and unknown general convex stage costs, and the resulting algorithms come with robust guarantees and tractability. However, for state-coupled storage dynamics, unconstrained DAC does not enforce per-step feasibility by construction. Our framework fills this gap by replacing DAC with SDAC, a simplex-constrained DAC-style family that preserves the feasibility defined by (1) while retaining the desirable characteristics.

D. Contributions

The contributions are organized in two layers:

- **Modeling and policy-class contribution.** We formulate adversarial online control of a storage system with decay-coupled state dynamics, state-dependent feasibility ($0 \leq u_t \leq \alpha x_t$), and state-dependent stage costs, and we introduce *Simplex Disturbance-Action Control* (SDAC), a DAC-inspired simplex-constrained policy class to compete with. The SDAC *enforces feasibility* by construction while preserving DAC-style expressiveness.
- **Algorithmic and analytical contribution.** We develop an exponentiated-gradient online controller over SDAC parameters in which a truncation step *enforces feasibility* of the online control action, and we analyze regret against the best fixed SDAC policy in hindsight, achieving sublinear regret with logarithmic dependence on policy dimension.

E. Paper Organization

Section III formalizes the adversarial storage-control model, assumptions, and regret benchmark. Section IV introduces the SDAC policy class and summarizes its structural properties. Section V presents the online controller, Section VI develops regret guarantees, and Section VII concludes. Deferred proofs appear in the Appendix.

II. NOTATION

We write $[z]_+ \triangleq \max\{z, 0\}$. For a vector $v \in \mathbb{R}^n$, the i -th coordinate is $v^{(i)}$, and $\mathbf{1}$ denotes the all-ones vector (dimension clear from context). The Euclidean inner product is $\langle \cdot, \cdot \rangle$. Time indices may be nonpositive, with the convention that the corresponding quantities are zero (in particular $w_\tau \equiv 0$ for $\tau \leq 0$). The online algorithm is denoted $\mathcal{A}$; a fixed policy is denoted π , or $\pi(M)$ when parameterized by parameters M .

III. MODEL AND PROBLEM SETUP

We consider a discrete-time process over times $t \in \{1, 2, \dots, T\}$. The system state is denoted by $x_t \in \mathbb{R}_{\geq 0}$, where x_t represents the resource level at the beginning of time t . The initial state is given as

$$x_1 = 0.$$

Fix a state-decay factor $\alpha \in (0, 1)$. At each time, the controller chooses a control action $u_t \in \mathbb{R}_{\geq 0}$ subject to the state constraint:

$$0 \leq u_t \leq \alpha x_t. \quad (1)$$

The system then receives an disturbance $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \quad (2)$$

After the control action is selected, a convex stage cost

$$c_t : \mathbb{R}_{\geq 0}^2 \rightarrow \mathbb{R}$$

is revealed, and the incurred cost is $c_t(x_t, u_t)$. We assume each c_t is convex in (x, u) and satisfies a Lipschitz condition on the relevant domain. There exist constants $L_x, L_u > 0$ such that for all admissible (x, u) and (x', u') ,

$$|c_t(x, u) - c_t(x', u')| \leq L_x |x - x'| + L_u |u - u'|. \quad (3)$$

The information pattern is adversarial and non-anticipatory. At time t , the controller chooses u_t before observing w_t and c_t . Hence both the disturbance sequence $\{w_t\}_{t=1}^T$ and the stage-cost sequence $\{c_t\}_{t=1}^T$ may be arbitrary.

a) Assumptions: Equations (1)–(3) define the model. For analysis, we collect the remaining standing assumptions as follows:

- 1) **Decay factor:** $\alpha \in (0, 1)$ is fixed and known.
- 2) **Disturbance bounds:** $w_t \in [0, 1]$ for all t .
- 3) **Adversarial timing:** at time t , u_t is chosen before observing w_t and c_t .
- 4) **Cost regularity:** each c_t is convex in (x, u) and obeys (3).

- 5) **Feasible policy class:** both the online policy and benchmark policies are required to satisfy (1) under dynamics (2).

For an online algorithm $\mathcal{A}$ producing trajectory $\{(x_t^{\mathcal{A}}, u_t^{\mathcal{A}})\}_{t=1}^T$, define cumulative cost

$$J_T(\mathcal{A}) \triangleq \sum_{t=1}^T c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}). \quad (4)$$

For fixed H , let the policy class be $\Pi \triangleq \text{SDAC}_H$ (defined in Section IV). For each $\pi \in \Pi$, let $\{(x_t^\pi, u_t^\pi)\}_{t=1}^T$ be the induced feasible trajectory under the same disturbance sequence. Regret is defined by

$$\text{Regret}(T) \triangleq \sum_{t=1}^T c_t(x_t^{\mathcal{A}}, u_t^{\mathcal{A}}) - \min_{\pi \in \Pi} \sum_{t=1}^T c_t(x_t^\pi, u_t^\pi). \quad (5)$$

While the online controller may update its SDAC parameters over time, the benchmark in (5) is the best single fixed SDAC policy chosen in hindsight. The objective is to design an online policy that preserves feasibility in (1) for all t and achieves sublinear regret as defined in (5).

IV. SIMPLEX DISTURBANCE-ACTION CONTROL (SDAC)

A. From Disturbance-Action Control to SDAC

Disturbance-Action Control (DAC) parameterizes the control action as a linear function of a finite window of past disturbances. This representation is central in adversarial online control because the cost is convex in term of the parameters, yields robust regret guarantees, and enables efficient online convex-optimization updates. In our setting, a representative DAC form is

$$u_t = \sum_{i=1}^H M^{(i)} w_{t-i}, \quad (6)$$

where $H \in \mathbb{N}$ is the memory horizon and $M \in \mathbb{R}^H$ is a linear filter. We allow nonpositive indices in such sums, with the convention $w_\tau \equiv 0$ for $\tau \leq 0$.

For our constrained model, unconstrained DAC is insufficient. In particular, the control action can violate either bound, with $u_t > \alpha x_t$ or $u_t < 0$. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which replaces unconstrained DAC filters by a simplex-constrained family tailored to this dynamics. SDAC is designed to keep the DAC advantages above (adversarial robustness, regret-based comparison, and tractable online updates) while adding feasibility by construction.

B. SDAC Definition

Fix memory horizon $H \in \mathbb{N}$. Let

$$\Delta_H \triangleq \left\{ M \in \mathbb{R}_{\geq 0}^H : \sum_{i=1}^H M^{(i)} \leq 1 \right\} \quad (7)$$

be the sub-probability simplex. For any $M \in \Delta_H$, define the policy $\pi(M)$ by

$$u_t^{\pi(M)} \triangleq \sum_{i=1}^H M^{(i)} \alpha^i w_{t-i}, \quad w_\tau \equiv 0 \text{ for } \tau \leq 0. \quad (8)$$

For each time t , let

$$\tilde{w}_t \triangleq (\alpha w_{t-1}, \dots, \alpha^H w_{t-H})^\top \in \mathbb{R}^H,$$

so that (8) can be written as

$$u_t^{\pi(M)} = \langle M, \tilde{w}_t \rangle. \quad (9)$$

The induced state trajectory satisfies

$$x_{t+1}^{\pi(M)} = \alpha x_t^{\pi(M)} - u_t^{\pi(M)} + w_t, \quad x_1^{\pi(M)} = 0. \quad (10)$$

The weights α^i align the disturbance filter with the decay factor α in (2).

First, feasibility is preserved within the class: for any fixed $M \in \Delta_H$, the induced trajectory under (8)–(10) satisfies $0 \leq u_t^{\pi(M)} \leq \alpha x_t^{\pi(M)}$ for all t (see Lemma 4).

Second, SDAC is compatible with online convex optimization. For fixed disturbance sequences, $u_t^{\pi(M)}$ is linear in M and $x_t^{\pi(M)}$ is affine in M (see Lemma 2). Therefore, since each c_t is convex in (x, u) , the induced cost is convex in M (see Lemma 5). Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

C. Expressiveness: Approximating Linear State-Feedback Policies

A standard policy class in online control is the family of linear state-feedback policies $u_t = -Kx_t$, and a foundational property of Disturbance-Action Control is that the DAC class approximates any strongly stable linear policy, with approximation error decaying geometrically in the memory horizon H [13], [17].

We establish the analogue of this guarantee for SDAC. In our scalar, state-constrained model the feasible linear state-feedback policies are exactly the proportional (constant-fraction) control actions $u_t = q x_t$ with gain $q \in [0, \alpha]$: since $x_t \geq 0$, the feasibility constraint $0 \leq u_t \leq \alpha x_t$ in (1) holds if and only if $q \in [0, \alpha]$. The next lemma shows that every such policy with $q \in (0, \alpha)$ is approximated, *within the feasible class* SDAC_H , by explicit parameters, with control-action, state, and cost errors that decay geometrically in H .

Lemma 1 (SDAC approximates linear state-feedback policies). *Fix $q \in (0, \alpha)$, set $\beta \triangleq \alpha - q$, and let π_q^{lin} denote the feasible linear policy*

$$u_t^{\text{lin}} = q x_t^{\text{lin}}, \quad x_{t+1}^{\text{lin}} = \beta x_t^{\text{lin}} + w_t, \quad x_1^{\text{lin}} = 0.$$

Define $M_q^* \in \mathbb{R}^H$ by

$$M_q^{*(i)} \triangleq \frac{q}{\alpha} \left(\frac{\beta}{\alpha} \right)^{i-1}, \quad i = 1, \dots, H.$$

Then $M_q^* \in \Delta_H$ with $\sum_{i=1}^H M_q^{*(i)} = 1 - (\beta/\alpha)^H$, and for every $\{w_t\} \subset [0, 1]$ and all $t \geq 1$,

$$\begin{aligned} 0 &\leq u_t^{\text{lin}} - u_t^{\pi(M_q^*)} \leq \frac{q \beta^H}{1 - \beta}, \\ 0 &\leq x_t^{\pi(M_q^*)} - x_t^{\text{lin}} \leq \frac{q \beta^H}{(1 - \beta)(1 - \alpha)}. \end{aligned} \quad (11)$$

Moreover, if each c_t satisfies (3), then

$$\begin{aligned} &\left| \sum_{t=1}^T c_t(x_t^{\pi(M_q^*)}, u_t^{\pi(M_q^*)}) - \sum_{t=1}^T c_t(x_t^{\text{lin}}, u_t^{\text{lin}}) \right| \\ &\leq \left(L_u + \frac{L_x}{1 - \alpha} \right) \frac{q \beta^H}{1 - \beta} T. \end{aligned} \quad (12)$$

In particular, the errors decay geometrically in H , so SDAC_H approximates π_q^{lin} arbitrarily well as $H \rightarrow \infty$. The proof is deferred to Appendix A.

This expressiveness property has a direct algorithmic consequence. Because the regret bound of Theorem 1 depends on the memory horizon only through $\sqrt{\log(H+1)}$, the window H can be enlarged cheaply; combining Lemma 1 with that bound then shows the online controller competes with the best fixed linear policy. We make this precise in Corollary 1, mirroring the role of DAC-to-linear approximation in nonstochastic control [13].

V. ONLINE ALGORITHM

We now present our online algorithm, which updates SDAC parameters by unnormalized-negentropy online mirror descent (equivalently, exponentiated gradient) over the sub-probability simplex $\Delta_H \subset \mathbb{R}^H$.

For each time t , let $M_t \in \Delta_H$ be the online parameter vector. We initialize at

$$M_1^{(i)} \triangleq \frac{1}{H+1}, \quad i = 1, \dots, H \quad (13)$$

so that the unnormalized-negentropy Bregman diameter of Δ_H relative to M_1 is at most $\log(H+1)$ (see the proof of Theorem 1).

a) *Control-action selection*: The controller first forms the unconstrained SDAC control action

$$\hat{u}_t \triangleq \langle M_t, \tilde{w}_t \rangle, \quad (14)$$

then enforces feasibility by truncation:

$$u_t^A \triangleq \min\{\hat{u}_t, \alpha x_t^A\}. \quad (15)$$

The state update is

$$x_{t+1}^A = \alpha x_t^A - u_t^A + w_t, \quad x_1^A = 0. \quad (16)$$

b) *Feasibility of the online trajectory*: We verify that the algorithm satisfies (1) at every time. The key mechanism is the truncation step in (15), which enforces per-step feasibility by clipping the unconstrained control action to the currently available feasible bound αx_t^A . Since $M_t \in \Delta_H$ and $\tilde{w}_t \geq 0$ componentwise, equation (14) gives $\hat{u}_t \geq 0$. By (15),

$$u_t^A = \min\{\hat{u}_t, \alpha x_t^A\} \in [0, \alpha x_t^A],$$

so the control-action feasibility constraint holds whenever $x_t^A \geq 0$. It remains to prove $x_t^A \geq 0$ for all t . The base case is $x_1^A = 0$. If $x_t^A \geq 0$, then $u_t^A \leq \alpha x_t^A$ and $w_t \in [0, 1]$, so from (16),

$$x_{t+1}^A = \alpha x_t^A - u_t^A + w_t \geq 0.$$

Thus, by induction, $x_t^A \geq 0$ and $0 \leq u_t^A \leq \alpha x_t^A$ for all t .

c) *Surrogate cost and gradient step*: Define the surrogate cost at time t by

$$\ell_t(M) \triangleq c_t(x_t^{\pi(M)}, u_t^{\pi(M)}), \quad M \in \Delta_H, \quad (17)$$

which is convex and Lipschitz by Lemma 5. Let $g_t \in \partial \ell_t(M_t)$ be a subgradient.

The exponentiated-gradient update is

$$\widetilde{M}_{t+1}^{(i)} \triangleq M_t^{(i)} \exp(-\eta g_t^{(i)}), \quad i = 1, \dots, H, \quad (18)$$

followed by the closed-form Bregman projection onto Δ_H ¹:

$$M_{t+1}^{(i)} \triangleq \frac{\widetilde{M}_{t+1}^{(i)}}{\max(1, S_{t+1})}, \quad S_{t+1} \triangleq \sum_{j=1}^H \widetilde{M}_{t+1}^{(j)}, \quad (19)$$

for $i = 1, \dots, H$.

This yields a feasible online trajectory satisfying (1) by (15) and performs online convex optimization over Δ_H through (17)–(19).

VI. REGRET ANALYSIS

The regret proof has two parts: a standard online convex optimization bound for the surrogate costs ℓ_t , and a stability bound showing that the truncated online control action stays close to the SDAC policy trajectory driven by the same parameters. The technical lemmas are deferred to Appendix A.

Theorem 1 (Regret bound). *Suppose each cost function c_t is convex and satisfies (3). Then the truncated exponentiated-gradient controller from Section V, initialized as in (13), satisfies*

$$\text{Regret}(T) \leq \frac{\log(H+1)}{\eta} + \eta L^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right),$$

$$L := \alpha L_u + \frac{L_x}{1-\alpha}.$$

In particular, the choice

$$\eta^* = \sqrt{\frac{\log(H+1)}{L^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right)}}$$

gives

$$\text{Regret}(T) \leq 2L \sqrt{\left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right) T \log(H+1)}.$$

Proof sketch. By Lemma 5, the surrogate costs are convex and L -Lipschitz. Unnormalized-negentropy OMD on Δ_H , initialized as in (13), yields Bregman diameter at most $\log(H+1)$, so the OMD term is bounded by $\log(H+1)/\eta + \eta L^2 T/2$. The remaining stability term is controlled by Lemmas 7–13, which together contribute at most $\frac{\alpha}{(1-\alpha)^2} \eta L^2 T$. Combining the two bounds yields the claim. $\square$

Combining Theorem 1 with the approximation guarantee of Lemma 1 shows that the controller is competitive not only with the best fixed SDAC policy but with the entire family of feasible linear state-feedback policies.

¹This step is exactly the Bregman (generalized-KL) projection of $\widetilde{M}_{t+1}$ onto $\Delta_H = \{M \geq 0 : \sum_i M^{(i)} \leq 1\}$ induced by the unnormalized negentropy; renormalization is triggered only when the unprojected mirror step leaves Δ_H , i.e. when $\sum_j \widetilde{M}_{t+1}^{(j)} > 1$. See Lemma 6.

Corollary 1 (No-regret against linear state-feedback policies). *Run the truncated exponentiated-gradient controller of Section V with the step size η^* of Theorem 1. Fix any gain $q \in (0, \alpha)$, set $\beta \triangleq \alpha - q$, and let $(x_t^{\text{lin}}, u_t^{\text{lin}})$ be the trajectory of the linear policy π_q^{lin} from Lemma 1. Then*

$$\begin{aligned} & \sum_{t=1}^T c_t(x_t^A, u_t^A) - \sum_{t=1}^T c_t(x_t^{\text{lin}}, u_t^{\text{lin}}) \\ & \leq 2L \sqrt{\left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right) T \log(H+1)} \\ & \quad + \left(L_u + \frac{L_x}{1-\alpha} \right) \frac{q \beta^H}{1-\beta} T, \end{aligned}$$

where $L = \alpha L_u + \frac{L_x}{1-\alpha}$. In particular, choosing the memory horizon $H = \lceil \log T / \log(1/\beta) \rceil$ forces $\beta^H \leq 1/T$, so the second term is $O(1)$ and

$$\begin{aligned} & \sum_{t=1}^T c_t(x_t^A, u_t^A) - \sum_{t=1}^T c_t(x_t^{\text{lin}}, u_t^{\text{lin}}) \\ & = O\left(\sqrt{T \log \log T}\right). \end{aligned}$$

Thus the controller is no-regret against every fixed linear state-feedback policy $u_t = q x_t$, $q \in (0, \alpha)$.

Proof. By the regret definition (5) and the optimized bound of Theorem 1,

$$\begin{aligned} \sum_{t=1}^T c_t(x_t^A, u_t^A) & \leq \min_{M \in \Delta_H} \sum_{t=1}^T c_t(x_t^{\pi(M)}, u_t^{\pi(M)}) \\ & \quad + 2L \sqrt{\left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right) T \log(H+1)}. \end{aligned}$$

Since $M_q^* \in \Delta_H$ by Lemma 1, the minimum is no larger than the cost of $\pi(M_q^*)$, which by (12) obeys

$$\begin{aligned} & \sum_{t=1}^T c_t(x_t^{\pi(M_q^*)}, u_t^{\pi(M_q^*)}) \\ & \leq \sum_{t=1}^T c_t(x_t^{\text{lin}}, u_t^{\text{lin}}) + \left(L_u + \frac{L_x}{1-\alpha} \right) \frac{q \beta^H}{1-\beta} T. \end{aligned}$$

Chaining the two displays gives the stated bound. For the asymptotic statement, $H = \lceil \log T / \log(1/\beta) \rceil$ gives $\beta^H \leq 1/T$, so the second term is at most $(L_u + \frac{L_x}{1-\alpha}) \frac{q}{1-\beta} = O(1)$, while $\log(H+1) = O(\log \log T)$ in the first term. $\square$

VII. CONCLUSION

We studied adversarial online control of a storage system under decay-coupled state dynamics and state-dependent feasibility constraints. The SDAC policy class enforces feasibility by construction, and the truncated exponentiated-gradient controller over SDAC parameters achieves sublinear regret against the best fixed SDAC policy, with only logarithmic dependence on the memory horizon. The same guarantees extend to competition with feasible linear state-feedback policies via the SDAC approximation result. The framework is directly applicable to energy-harvesting battery management and related storage settings in which decisions must respect state constraints in real time.

APPENDIX A DEFERRED PROOFS

This appendix collects the deferred proofs for structural SDAC properties used in Section IV and the regret bound stated in Section VI.

Recall that we set $w_\tau \equiv 0$ for $\tau \leq 0$, which is used in several base-case arguments below.

We first prove the linear-policy approximation guarantee stated in Section IV-C.

Proof of Lemma 1. Fix $q \in (0, \alpha)$, set $\beta = \alpha - q \in (0, \alpha)$, and recall $w_\tau \equiv 0$ for $\tau \leq 0$.

Feasibility and closed form of the linear policy. Substituting $u_t^{\text{lin}} = q x_t^{\text{lin}}$ into its state update gives the closed loop $x_{t+1}^{\text{lin}} = \alpha x_t^{\text{lin}} - q x_t^{\text{lin}} + w_t = \beta x_t^{\text{lin}} + w_t$ with $x_1^{\text{lin}} = 0$. Since $\beta > 0$ and $w_t \geq 0$, induction yields $x_t^{\text{lin}} \geq 0$, and then $0 \leq u_t^{\text{lin}} = q x_t^{\text{lin}} \leq \alpha x_t^{\text{lin}}$ because $q \leq \alpha$; thus π_q^{lin} satisfies (1). Unrolling the closed-loop recursion from $x_1^{\text{lin}} = 0$,

$$x_t^{\text{lin}} = \sum_{j=1}^{t-1} \beta^{j-1} w_{t-j}, \quad u_t^{\text{lin}} = q \sum_{j=1}^{t-1} \beta^{j-1} w_{t-j}.$$

Part 1. Each $M_q^{*(i)} = \frac{q}{\alpha} (\beta/\alpha)^{i-1} \geq 0$. Using $1 - \beta/\alpha = (\alpha - \beta)/\alpha = q/\alpha$ and the finite geometric sum,

$$\begin{aligned} \sum_{i=1}^H M_q^{*(i)} &= \frac{q}{\alpha} \cdot \frac{1 - (\beta/\alpha)^H}{1 - \beta/\alpha} \\ &= \frac{q}{\alpha} \cdot \frac{\alpha}{q} (1 - (\beta/\alpha)^H) = 1 - (\beta/\alpha)^H < 1, \end{aligned}$$

since $0 < \beta/\alpha < 1$. Hence $M_q^* \in \Delta_H$ and $\pi(M_q^*) \in \text{SDAC}_H$.

Part 2. By the SDAC definition (8) and the identity $M_q^{*(i)} \alpha^i = q \beta^{i-1}$,

$$u_t^{\pi(M_q^*)} = \sum_{i=1}^H M_q^{*(i)} \alpha^i w_{t-i} = q \sum_{i=1}^H \beta^{i-1} w_{t-i}.$$

Because $w_{t-i} = 0$ for $i \geq t$, this equals $q \sum_{i=1}^{\min\{H, t-1\}} \beta^{i-1} w_{t-i}$, so subtracting it from u_t^{lin} cancels the first $\min\{H, t-1\}$ terms:

$$u_t^{\text{lin}} - u_t^{\pi(M_q^*)} = q \sum_{j=H+1}^{t-1} \beta^{j-1} w_{t-j} \geq 0,$$

which equals 0 when $t \leq H+1$. Using $w_{t-j} \in [0, 1]$ and summing the geometric tail,

$$u_t^{\text{lin}} - u_t^{\pi(M_q^*)} \leq q \sum_{j=H+1}^{\infty} \beta^{j-1} = \frac{q \beta^H}{1 - \beta}.$$

Part 3. Let $\delta_t \triangleq x_t^{\pi(M_q^*)} - x_t^{\text{lin}}$ and $e_t \triangleq u_t^{\text{lin}} - u_t^{\pi(M_q^*)} \in [0, q \beta^H / (1 - \beta)]$ by Part 2. The two trajectories follow the same dynamics (2) driven by the same exogenous disturbance, so the exogenous disturbance cancels on subtraction:

$$\delta_{t+1} = \alpha \delta_t - (u_t^{\pi(M_q^*)} - u_t^{\text{lin}}) = \alpha \delta_t + e_t, \quad \delta_1 = 0.$$

Unrolling gives $\delta_t = \sum_{s=1}^{t-1} \alpha^{t-1-s} e_s \geq 0$, and since $e_s \leq q \beta^H / (1 - \beta)$ and $\sum_{s=1}^{t-1} \alpha^{t-1-s} \leq 1/(1 - \alpha)$,

$$0 \leq \delta_t \leq \frac{q \beta^H}{1 - \beta} \cdot \frac{1}{1 - \alpha} = \frac{q \beta^H}{(1 - \beta)(1 - \alpha)}.$$

Part 4. By the Lipschitz condition (3) and Parts 2–3, for each t ,

$$\begin{aligned} |c_t(x_t^{\pi(M_q^*)}, u_t^{\pi(M_q^*)}) - c_t(x_t^{\text{lin}}, u_t^{\text{lin}})| \\ \leq L_x |\delta_t| + L_u |e_t| \leq \left(\frac{L_x}{1 - \alpha} + L_u \right) \frac{q \beta^H}{1 - \beta}. \end{aligned}$$

Summing over $t = 1, \dots, T$ and applying the triangle inequality yields Part 4. $\square$

Lemma 2 (State decomposition under SDAC). *Fix $H \in \mathbb{N}$ and $M \in \Delta_H$, and write $W \triangleq \sum_{i=1}^H M^{(i)} \leq 1$. Let $\pi(M)$ be defined by (8) and (10). Then for all $t \geq 2$,*

$$x_t^{\pi(M)} = \sum_{i=1}^H M^{(i)} \psi_t^{(i)} + (1 - W) \rho_t = \langle M, \psi_t \rangle + (1 - W) \rho_t, \quad (20)$$

where

$$\psi_t^{(i)} \triangleq \sum_{j=1}^i \alpha^{j-1} w_{t-j} \quad (i = 1, \dots, H),$$

and

$$\rho_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j}.$$

Equivalently, ρ_t is the state $x_t^{\pi(0)}$ produced by the null policy $M = 0$ (release nothing, $x_{\tau+1} = \alpha x_\tau + w_\tau$), and the coefficients satisfy $0 \leq \psi_t^{(i)} \leq \rho_t \leq 1/(1 - \alpha)$. Consequently, setting $M^{(0)} \triangleq 1 - W \geq 0$, the identity (20) writes $x_t^{\pi(M)}$ as the convex combination $M^{(0)} \rho_t + \sum_{i=1}^H M^{(i)} \psi_t^{(i)}$ of quantities in $[0, 1/(1 - \alpha)]$; in particular $M \mapsto x_t^{\pi(M)}$ is affine and

$$0 \leq x_t^{\pi(M)} \leq \frac{1}{1 - \alpha} \quad \text{for all } t \geq 2.$$

Proof. Recall $w_\tau \equiv 0$ for $\tau \leq 0$. Isolating the $j = 1$ term in the definition of ρ_{t+1} and reindexing gives the one-step identity $\rho_{t+1} = \alpha \rho_t + w_t$ (with $\rho_1 = 0$). We prove (20) by induction on t .

Base case $t = 2$. Since $x_1^{\pi(M)} = 0$ and $w_{1-i} = 0$ for all $i \geq 1$, the policy (8) gives $u_1^{\pi(M)} = \sum_{i=1}^H M^{(i)} \alpha^i w_{1-i} = 0$, and the update (10) gives $x_2^{\pi(M)} = \alpha x_1^{\pi(M)} - u_1^{\pi(M)} + w_1 = w_1$. On the other hand $\psi_2^{(i)} = w_1$ for every i and $\rho_2 = w_1$, so $\langle M, \psi_2 \rangle + (1 - W) \rho_2 = W w_1 + (1 - W) w_1 = w_1$, matching (20).

Inductive step. Assume (20) holds at time $t \geq 2$. Using (8), (10), and the induction hypothesis,

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \alpha x_t^{\pi(M)} - \sum_{i=1}^H M^{(i)} \alpha^i w_{t-i} + w_t \\ &= \left(\alpha \langle M, \psi_t \rangle - \sum_{i=1}^H M^{(i)} \alpha^i w_{t-i} \right) \\ &\quad + \alpha(1 - W) \rho_t + w_t. \end{aligned}$$

Reindexing the inner summation ($j \mapsto j + 1$) yields

$$\begin{aligned} \alpha \langle M, \psi_t \rangle - \sum_{i=1}^H M^{(i)} \alpha^i w_{t-i} \\ &= \sum_{i=1}^H M^{(i)} \left(\sum_{j=1}^{i-1} \alpha^j w_{t-j} \right) \\ &= \sum_{i=1}^H M^{(i)} (\psi_{t+1}^{(i)} - w_t) = \langle M, \psi_{t+1} \rangle - W w_t, \end{aligned}$$

where the second equality uses $\psi_{t+1}^{(i)} = w_t + \sum_{j=1}^{i-1} \alpha^j w_{t-j}$. Substituting this together with $\rho_{t+1} = \alpha \rho_t + w_t$,

$$\begin{aligned} x_{t+1}^{\pi(M)} &= \langle M, \psi_{t+1} \rangle - W w_t + \alpha(1 - W) \rho_t + w_t \\ &= \langle M, \psi_{t+1} \rangle + (1 - W)(w_t + \alpha \rho_t) \\ &= \langle M, \psi_{t+1} \rangle + (1 - W) \rho_{t+1}, \end{aligned}$$

which is (20) at time $t + 1$.

Bounds. Each $\psi_t^{(i)}$ is the truncation to $j \leq i$ of the nonnegative series defining ρ_t , so $0 \leq \psi_t^{(i)} \leq \rho_t$; and since $w_t \in [0, 1]$, the geometric series gives $\rho_t = \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j} \leq \sum_{j=1}^{\infty} \alpha^{j-1} = 1/(1 - \alpha)$. With $M^{(0)} = 1 - W \geq 0$, the parameters $M^{(0)}, M^{(1)}, \dots, M^{(H)}$ are nonnegative and sum to one, so (20) exhibits $x_t^{\pi(M)}$ as a convex combination of $\rho_t, \psi_t^{(1)}, \dots, \psi_t^{(H)} \in [0, 1/(1 - \alpha)]$, giving $0 \leq x_t^{\pi(M)} \leq 1/(1 - \alpha)$. Affineness in M is immediate, as ψ_t and ρ_t do not depend on M . $\square$

Lemma 3 (Linearity of SDAC action in M). *Under the setup of Lemma 2, for each t ,*

$$u_t^{\pi(M)} = \langle M, \tilde{w}_t \rangle,$$

where $\tilde{w}_t$ is the decayed past-disturbance vector from (9), with components

$$\tilde{w}_t^{(i)} \triangleq \alpha^i w_{t-i}.$$

Moreover, $0 \leq \tilde{w}_t^{(i)} \leq \alpha$ for all i .

Proof. From (8),

$$u_t^{\pi(M)} = \sum_{i=1}^H M^{(i)} \alpha^i w_{t-i} = \sum_{i=1}^H M^{(i)} \tilde{w}_t^{(i)} = \langle M, \tilde{w}_t \rangle.$$

The lower bound is immediate from $w_{t-i} \geq 0$. For the upper bound, use $w_t \in [0, 1]$:

$$\tilde{w}_t^{(i)} = \alpha^i w_{t-i} \leq \alpha^i \leq \alpha.$$

$\square$

Lemma 4 (Feasibility of SDAC policies). *For every fixed $M \in \Delta_H$, the trajectory induced by (8)–(10) satisfies*

$$0 \leq u_t^{\pi(M)} \leq \alpha x_t^{\pi(M)} \quad \text{for all } t \geq 1.$$

Proof. Nonnegativity of $u_t^{\pi(M)}$ follows from (8), because $M^{(i)} \geq 0$, $\alpha^i \geq 0$, and $w_{t-i} \geq 0$ for every i .

For the upper bound at $t = 1$, we have $x_1^{\pi(M)} = 0$ and $w_{1-i} = 0$ for all $i \geq 1$, so $u_1^{\pi(M)} = 0 = \alpha x_1^{\pi(M)}$.

Now fix $t \geq 2$. Lemma 2 gives $x_t^{\pi(M)} = \langle M, \psi_t \rangle + (1 - W) \rho_t$ with $W = \sum_{i=1}^H M^{(i)} \leq 1$. Since $(1 - W) \rho_t \geq 0$,

$$\begin{aligned} \alpha x_t^{\pi(M)} &\geq \alpha \langle M, \psi_t \rangle = \alpha \sum_{i=1}^H M^{(i)} \left(\sum_{j=1}^i \alpha^{j-1} w_{t-j} \right) \\ &= \sum_{i=1}^H M^{(i)} \left(\sum_{j=1}^i \alpha^j w_{t-j} \right). \end{aligned}$$

Collecting the coefficient of each w_{t-i} and using that all parameters are nonnegative, for every i we have $\sum_{j=i}^H M^{(j)} \geq M^{(i)}$, so

$$\begin{aligned} \alpha x_t^{\pi(M)} &\geq \sum_{i=1}^H \left(\sum_{j=i}^H M^{(j)} \right) \alpha^i w_{t-i} \\ &\geq \sum_{i=1}^H M^{(i)} \alpha^i w_{t-i} = u_t^{\pi(M)}, \end{aligned}$$

since $w_{t-i} \geq 0$. $\square$

Lemma 5 (Convexity and Lipschitz continuity of surrogate cost). *Define*

$$\ell_t(M) \triangleq c_t(x_t^{\pi(M)}, u_t^{\pi(M)}), \quad M \in \Delta_H.$$

Then ℓ_t is convex on Δ_H , and for all $M, M' \in \Delta_H$,

$$|\ell_t(M) - \ell_t(M')| \leq \left(\frac{L_x}{1 - \alpha} + \alpha L_u \right) \|M - M'\|_1.$$

Proof. By Lemma 2 and Lemma 3, with $W = \sum_{i=1}^H M^{(i)}$,

$$\ell_t(M) = c_t(\langle M, \psi_t \rangle + (1 - W) \rho_t, \langle M, \tilde{w}_t \rangle).$$

The map $M \mapsto (x_t^{\pi(M)}, u_t^{\pi(M)})$ is affine (Lemmas 2–3) and c_t is convex, so ℓ_t is convex.

For the Lipschitz estimate, the decomposition gives, for all $M, M' \in \Delta_H$,

$$\begin{aligned} x_t^{\pi(M)} - x_t^{\pi(M')} &= \langle M - M', \psi_t \rangle - (W - W') \rho_t \\ &= \langle M - M', \psi_t - \rho_t \mathbf{1} \rangle, \end{aligned}$$

using $W - W' = \langle M - M', \mathbf{1} \rangle$, while $u_t^{\pi(M)} - u_t^{\pi(M')} = \langle M - M', \tilde{w}_t \rangle$. Hence, by (3) and Hölder's inequality,

$$\begin{aligned} |\ell_t(M) - \ell_t(M')| &\leq L_x |\langle M - M', \psi_t - \rho_t \mathbf{1} \rangle| \\ &\quad + L_u |\langle M - M', \tilde{w}_t \rangle| \\ &\leq (L_x \|\psi_t - \rho_t \mathbf{1}\|_\infty + L_u \|\tilde{w}_t\|_\infty) \\ &\quad \cdot \|M - M'\|_1. \end{aligned}$$

Since $0 \leq \psi_t^{(i)} \leq \rho_t \leq 1/(1 - \alpha)$ (Lemma 2), we have $\|\psi_t - \rho_t \mathbf{1}\|_\infty = \max_i (\rho_t - \psi_t^{(i)}) \leq \rho_t \leq 1/(1 - \alpha)$, and Lemma 3 gives $\|\tilde{w}_t\|_\infty \leq \alpha$. This yields the claim, and one may take the surrogate Lipschitz constant

$$L := \alpha L_u + \frac{L_x}{1 - \alpha},$$

as used in the main text. $\square$

Lemma 6 (Conditional normalization is the KL projection onto Δ_H). *Let $\Psi(M) = \sum_{i=1}^H (M^{(i)} \log M^{(i)} - M^{(i)})$ denote*

the unnormalized negentropy on $\mathbb{R}_{\geq 0}^H$ (with the convention $0 \log 0 = 0$), and let

$$D_\Psi(M \| V) = \sum_{i=1}^H \left(M^{(i)} \log \frac{M^{(i)}}{V^{(i)}} - M^{(i)} + V^{(i)} \right)$$

be its Bregman divergence, the generalized Kullback–Leibler divergence. Fix $\widetilde{M} \in \mathbb{R}_{>0}^H$ and set $S \triangleq \sum_{i=1}^H \widetilde{M}^{(i)}$. Then the Bregman projection of $\widetilde{M}$ onto $\Delta_H = \{M \in \mathbb{R}_{\geq 0}^H : \sum_i M^{(i)} \leq 1\}$,

$$P(\widetilde{M}) \triangleq \arg \min_{M \in \Delta_H} D_\Psi(M \| \widetilde{M}),$$

is given coordinatewise by

$$P(\widetilde{M})^{(i)} = \frac{\widetilde{M}^{(i)}}{\max(1, S)}.$$

Consequently, taking $\widetilde{M} = \widetilde{M}_{t+1}$ from (18), the update (19) is exactly the projection $M_{t+1} = P(\widetilde{M}_{t+1})$, and hence $M_{t+1} \in \Delta_H$.

Proof. Since $\widetilde{M}$ is fixed, the terms $+\widetilde{M}^{(i)}$ are constant, so $P(\widetilde{M})$ minimizes

$$F(M) \triangleq \sum_{i=1}^H \left(M^{(i)} \log \frac{M^{(i)}}{\widetilde{M}^{(i)}} - M^{(i)} \right)$$

over Δ_H . The function F is strictly convex on $\mathbb{R}_{>0}^H$ and Δ_H is closed and convex, so the minimizer is unique; its partial derivatives are $\partial F / \partial M^{(i)} = \log(M^{(i)} / \widetilde{M}^{(i)})$.

Case $S \leq 1$. Here $\widetilde{M} \in \Delta_H$. Setting $\partial F / \partial M^{(i)} = 0$ gives the unconstrained minimizer $M = \widetilde{M}$ over $\mathbb{R}_{\geq 0}^H$, which is feasible; hence $P(\widetilde{M}) = \widetilde{M}$.

Case $S > 1$. Here $\widetilde{M} \notin \Delta_H$, so the constraint $\sum_i M^{(i)} \leq 1$ must be active at the optimum. As $\widetilde{M}^{(i)} > 0$, the optimal coordinates are strictly positive and the nonnegativity constraints are inactive; introducing a multiplier λ for $\sum_i M^{(i)} = 1$, KKT stationarity reads

$$\log(M^{(i)} / \widetilde{M}^{(i)}) + \lambda = 0 \implies M^{(i)} = \widetilde{M}^{(i)} e^{-\lambda}.$$

The active constraint $\sum_i M^{(i)} = 1$ forces $e^{-\lambda} = 1/S$, so $M^{(i)} = \widetilde{M}^{(i)} / S$; moreover $\lambda = \log S > 0$, so dual feasibility holds. Equivalently, $P(\widetilde{M})^{(i)} = \widetilde{M}^{(i)} / \max(1, S)$ in both cases. $\square$

Lemma 7 (One-step movement of unnormalized-negentropy OMD). *For the unnormalized-negentropy OMD update with step size $\eta > 0$ and subgradient $g_t \in \partial \ell_t(M_t)$,*

$$\|M_{t+1} - M_t\|_1 \leq \eta \|g_t\|_\infty.$$

In particular, if $\|g_t\|_\infty \leq L$, then $\|M_{t+1} - M_t\|_1 \leq \eta L$.

Proof. Let $\Psi(M) = \sum_{i=1}^H (M^{(i)} \log M^{(i)} - M^{(i)})$. By Lemma 6, M_{t+1} is the Bregman projection of the mirror step $\widetilde{M}_{t+1}$ onto Δ_H , so the first order optimality condition gives, for all $M \in \Delta_H$,

$$\left\langle g_t + \frac{1}{\eta} (\nabla \Psi(M_{t+1}) - \nabla \Psi(M_t)), M - M_{t+1} \right\rangle \geq 0.$$

Taking $M = M_t$ and rearranging yields

$$\langle g_t, M_t - M_{t+1} \rangle \geq \frac{1}{\eta} \langle \nabla \Psi(M_{t+1}) - \nabla \Psi(M_t), M_{t+1} - M_t \rangle.$$

By the Bregman identity,

$$\begin{aligned} & \langle \nabla \Psi(M_{t+1}) - \nabla \Psi(M_t), M_{t+1} - M_t \rangle \\ &= D_\Psi(M_{t+1} \| M_t) + D_\Psi(M_t \| M_{t+1}). \end{aligned}$$

Hence, using Hölder's inequality,

$$\begin{aligned} & D_\Psi(M_{t+1} \| M_t) + D_\Psi(M_t \| M_{t+1}) \\ & \leq \eta \|g_t\|_\infty \|M_{t+1} - M_t\|_1. \end{aligned}$$

Since both iterates lie in Δ_H , the unnormalized negentropy is 1-strongly convex with respect to $\|\cdot\|_1$ on this set: the Hessian $\nabla^2 \Psi(M) = \text{diag}(1/M^{(i)})$ satisfies $v^\top \nabla^2 \Psi(M) v = \sum_i (v^{(i)})^2 / M^{(i)} \geq \|v\|_1^2 / (\sum_i M^{(i)}) \geq \|v\|_1^2$ whenever $\sum_i M^{(i)} \leq 1$ (Cauchy–Schwarz), so the usual Pinsker-type bound for the generalized KL extends to the sub-probability simplex Δ_H ,

$$\begin{aligned} D_\Psi(M_{t+1} \| M_t) & \geq \frac{1}{2} \|M_{t+1} - M_t\|_1^2, \\ D_\Psi(M_t \| M_{t+1}) & \geq \frac{1}{2} \|M_{t+1} - M_t\|_1^2, \end{aligned}$$

and the left-hand side is at least $\|M_{t+1} - M_t\|_1^2$. Therefore

$$\|M_{t+1} - M_t\|_1^2 \leq \eta \|g_t\|_\infty \|M_{t+1} - M_t\|_1,$$

which implies $\|M_{t+1} - M_t\|_1 \leq \eta \|g_t\|_\infty$. $\square$

Lemma 8 (Cumulative movement). *For every $t \geq 2$ and every integer i with $1 \leq i \leq t-1$,*

$$\|M_{t-i} - M_t\|_1 \leq i \eta L.$$

Proof. The range $1 \leq i \leq t-1$ ensures that every index $t-j$ for $j = 1, \dots, i$ satisfies $t-j \geq 1$. By the triangle inequality and Lemma 7,

$$\|M_{t-i} - M_t\|_1 \leq \sum_{j=1}^i \|M_{t-j} - M_{t-j+1}\|_1 \leq i \eta L. \quad \square$$

Lemma 9 (Control-action bounds relative to the benchmark trajectory). *For all t ,*

$$u_t^A \leq \langle M_t, \widetilde{w}_t \rangle - \alpha [x_t^{\pi(M_t)} - x_t^A]_+, \quad (21)$$

$$u_t^A \leq \langle M_t, \widetilde{w}_t \rangle. \quad (22)$$

Proof. Recall from (14)–(15) that

$$\widehat{u}_t = \langle M_t, \widetilde{w}_t \rangle, \quad u_t^A = \min\{\widehat{u}_t, \alpha x_t^A\}.$$

Therefore,

$$u_t^A = \widehat{u}_t - [\widehat{u}_t - \alpha x_t^A]_+ = \langle M_t, \widetilde{w}_t \rangle - [\langle M_t, \widetilde{w}_t \rangle - \alpha x_t^A]_+.$$

This identity gives the upper bound immediately, since the subtracted term is nonnegative.

For the lower bound, Lemma 4 applied to the benchmark policy $\pi(M_t)$ yields, using (9),

$$u_t^{\pi(M_t)} = \langle M_t, \widetilde{w}_t \rangle \leq \alpha x_t^{\pi(M_t)}.$$

Subtracting αx_t^A from both sides and applying monotonicity of $[\cdot]_+$ gives

$$\begin{aligned} & [\langle M_t, \tilde{w}_t \rangle - \alpha x_t^A]_+ \\ & \leq [\alpha x_t^{\pi(M_t)} - \alpha x_t^A]_+ = \alpha [x_t^{\pi(M_t)} - x_t^A]_+. \end{aligned}$$

Substituting this bound into the expression above for u_t^A yields the stated lower bound. $\square$

Lemma 10 (State-gap recursion). *Define the state gap*

$$\Gamma_t^M \triangleq x_t^{\pi(M)} - x_t^A. \quad (23)$$

For any fixed $M \in \Delta_H$ and every $t \geq 1$ the following two-sided recursion holds:

$$\begin{aligned} \Gamma_{t+1}^M &= \alpha \Gamma_t^M - (u_t^{\pi(M)} - u_t^A) \\ &\geq \alpha \Gamma_t^M + \langle M_t - M, \tilde{w}_t \rangle - \alpha [\Gamma_t^{M_t}]_+, \end{aligned} \quad (24)$$

$$\Gamma_{t+1}^M \leq \alpha \Gamma_t^M + \langle M_t - M, \tilde{w}_t \rangle. \quad (25)$$

Proof. From the state updates (10) (for $\pi(M)$) and (16) (for the algorithm), the exogenous disturbance w_t cancels and we obtain the exact identity

$$\Gamma_{t+1}^M = x_{t+1}^{\pi(M)} - x_{t+1}^A = \alpha \Gamma_t^M - (u_t^{\pi(M)} - u_t^A). \quad (26)$$

Consequently, bounding Γ_{t+1}^M reduces to bounding the control-action difference $u_t^{\pi(M)} - u_t^A$.

Using the SDAC representation (9) for the benchmark control action and the bounds of Lemma 9 for the algorithm control action, we have

$$u_t^{\pi(M)} = \langle M, \tilde{w}_t \rangle,$$

and

$$\langle M_t, \tilde{w}_t \rangle - \alpha [\Gamma_t^{M_t}]_+ \leq u_t^A \leq \langle M_t, \tilde{w}_t \rangle.$$

Subtracting the upper bound on u_t^A from $u_t^{\pi(M)}$ yields the lower bound on $u_t^{\pi(M)} - u_t^A$:

$$u_t^{\pi(M)} - u_t^A \geq \langle M, \tilde{w}_t \rangle - \langle M_t, \tilde{w}_t \rangle = -\langle M_t - M, \tilde{w}_t \rangle.$$

Inserting this into the identity (26) gives

$$\Gamma_{t+1}^M \leq \alpha \Gamma_t^M + \langle M_t - M, \tilde{w}_t \rangle,$$

which is (25).

Similarly, using the lower bound on u_t^A produces an upper bound on $u_t^{\pi(M)} - u_t^A$:

$$\begin{aligned} u_t^{\pi(M)} - u_t^A &\leq \langle M, \tilde{w}_t \rangle - (\langle M_t, \tilde{w}_t \rangle - \alpha [\Gamma_t^{M_t}]_+) \\ &= -\langle M_t - M, \tilde{w}_t \rangle + \alpha [\Gamma_t^{M_t}]_+. \end{aligned}$$

Substituting this bound into the identity (26) yields

$$\Gamma_{t+1}^M \geq \alpha \Gamma_t^M + \langle M_t - M, \tilde{w}_t \rangle - \alpha [\Gamma_t^{M_t}]_+,$$

which is (24) and completes the proof. $\square$

Lemma 11 (State deviation bound). *For every $t \geq 1$, let the state gap be defined by (23). Then*

$$-\frac{\alpha \eta L}{(1-\alpha)^3} \leq \Gamma_t^{M_t} = x_t^{\pi(M_t)} - x_t^A \leq \frac{\alpha \eta L}{(1-\alpha)^2}.$$

Proof. From (25), for any fixed benchmark $M \in \Delta_H$ and every $s \geq 1$,

$$\Gamma_{s+1}^M \leq \alpha \Gamma_s^M + \langle M_s - M, \tilde{w}_s \rangle.$$

Applying the recursion repeatedly up to time t yields

$$\begin{aligned} \Gamma_t^M &\leq \alpha \Gamma_{t-1}^M + \langle M_{t-1} - M, \tilde{w}_{t-1} \rangle \\ &\leq \alpha^2 \Gamma_{t-2}^M + \alpha \langle M_{t-2} - M, \tilde{w}_{t-2} \rangle \\ &\quad + \langle M_{t-1} - M, \tilde{w}_{t-1} \rangle \\ &\vdots \\ &\leq \alpha^{t-1} \Gamma_1^M + \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M, \tilde{w}_s \rangle. \end{aligned}$$

Now set the benchmark equal to the online parameter $M = M_t$. Since $x_1^{\pi(M_t)} = x_1^A = 0$, we have $\Gamma_1^{M_t} = 0$, and therefore

$$\Gamma_t^{M_t} \leq \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M_t, \tilde{w}_s \rangle.$$

Note that $\|\tilde{w}_s\|_\infty = \max_{i \leq H} \alpha^i w_{s-i} \leq \alpha$ since $w_\tau \in [0, 1]$ and $i \geq 1$. Hence,

$$\begin{aligned} \Gamma_t^{M_t} &\leq \sum_{s=1}^{t-1} \alpha^{t-1-s} |\langle M_s - M_t, \tilde{w}_s \rangle| \\ &\leq \sum_{s=1}^{t-1} \alpha^{t-1-s} \|M_s - M_t\|_1 \|\tilde{w}_s\|_\infty \\ &\leq \alpha \eta L \sum_{s=1}^{t-1} \alpha^{t-1-s} (t-s) \\ &= \alpha \eta L \sum_{j=1}^{t-1} j \alpha^{j-1} \leq \alpha \eta L \sum_{j=1}^{\infty} j \alpha^{j-1} = \frac{\alpha \eta L}{(1-\alpha)^2}. \end{aligned}$$

Where we used the geometric-series identity

$$\sum_{j=1}^{\infty} j \alpha^{j-1} = \frac{1}{(1-\alpha)^2}.$$

Similarly, for the lower bound, unrolling (24) with the same fixed benchmark $M = M_t$ yields

$$\Gamma_t^{M_t} \geq \sum_{s=1}^{t-1} \alpha^{t-1-s} \langle M_s - M_t, \tilde{w}_s \rangle - \sum_{s=1}^{t-1} \alpha^{t-s} [\Gamma_s^{M_s}]_+.$$

The upper bound just proved applies to every earlier time s , so $[\Gamma_s^{M_s}]_+ \leq \alpha \eta L / (1-\alpha)^2$. Also, by Hölder's inequality and Lemma 8,

$$\begin{aligned} \Gamma_t^{M_t} &\geq -\alpha \eta L \sum_{s=1}^{t-1} \alpha^{t-1-s} (t-s) \\ &\quad - \frac{\alpha \eta L}{(1-\alpha)^2} \sum_{s=1}^{t-1} \alpha^{t-s} \\ &= -\alpha \eta L \sum_{j=1}^{t-1} j \alpha^{j-1} - \frac{\alpha \eta L}{(1-\alpha)^2} \sum_{j=1}^{t-1} \alpha^j. \end{aligned}$$

Extending both sums to infinity gives

$$\Gamma_t^{M_t} \geq -\alpha \eta L \sum_{j=1}^{\infty} j \alpha^{j-1} - \frac{\alpha \eta L}{(1-\alpha)^2} \sum_{j=1}^{\infty} \alpha^j.$$

Using the geometric-series identities

$$\sum_{j=1}^{\infty} j \alpha^{j-1} = \frac{1}{(1-\alpha)^2}, \quad \sum_{j=1}^{\infty} \alpha^j = \frac{\alpha}{1-\alpha},$$

we obtain

$$\Gamma_t^{M_t} \geq -\frac{\alpha \eta L}{(1-\alpha)^2} - \frac{\alpha^2 \eta L}{(1-\alpha)^3} = -\frac{\alpha \eta L}{(1-\alpha)^3}.$$

Hence, the claim. $\square$

Lemma 12 (Action-state gap). *Let the state gap be defined by (23). Then for all t ,*

$$0 \leq u_t^{\pi(M_t)} - u_t^A \leq \alpha [\Gamma_t^{M_t}]_+ \leq \frac{\alpha^2 \eta L}{(1-\alpha)^2}.$$

Proof. By (9) and (14), the benchmark control action equals the unconstrained control action, $u_t^{\pi(M_t)} = \langle M_t, \tilde{w}_t \rangle = \hat{u}_t$, while $u_t^A = \min\{\hat{u}_t, \alpha x_t^A\} \leq \hat{u}_t$ by (15). Hence the gap is nonnegative and

$$u_t^{\pi(M_t)} - u_t^A = \hat{u}_t - \min\{\hat{u}_t, \alpha x_t^A\} = [\langle M_t, \tilde{w}_t \rangle - \alpha x_t^A]_+.$$

As established in the proof of Lemma 9 (using feasibility of $\pi(M_t)$),

$$[\langle M_t, \tilde{w}_t \rangle - \alpha x_t^A]_+ \leq \alpha [x_t^{\pi(M_t)} - x_t^A]_+ = \alpha [\Gamma_t^{M_t}]_+,$$

which is the middle inequality. Finally, by Lemma 11, $[\Gamma_t^{M_t}]_+ \leq \frac{\alpha \eta L}{(1-\alpha)^2}$, so

$$\alpha [\Gamma_t^{M_t}]_+ \leq \frac{\alpha^2 \eta L}{(1-\alpha)^2},$$

which gives the rightmost inequality and completes the proof. $\square$

Lemma 13 (Stability term). *For all horizons T ,*

$$\sum_{t=1}^T \left(c_t(x_t^A, u_t^A) - c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \right) \leq \frac{\alpha T \eta L^2}{(1-\alpha)^2},$$

where $L = \alpha L_u + \frac{L_x}{1-\alpha}$.

Proof. By Lipschitz continuity of c_t (see equation (3)),

$$\begin{aligned} c_t(x_t^A, u_t^A) - c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ \leq L_u |u_t^A - u_t^{\pi(M_t)}| + L_x |x_t^A - x_t^{\pi(M_t)}|. \end{aligned}$$

From Lemma 12,

$$|u_t^A - u_t^{\pi(M_t)}| \leq \frac{\alpha^2 \eta L}{(1-\alpha)^2},$$

Also, by Lemma 11 and (23),

$$|x_t^A - x_t^{\pi(M_t)}| = |\Gamma_t^{M_t}| \leq \frac{\alpha \eta L}{(1-\alpha)^3}.$$

Therefore, for each $t \geq 1$,

$$\begin{aligned} c_t(x_t^A, u_t^A) - c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ \leq L_u \frac{\alpha^2 \eta L}{(1-\alpha)^2} + L_x \frac{\alpha \eta L}{(1-\alpha)^3}. \end{aligned}$$

Summing over $t = 1, 2, \dots, T$ gives

$$\begin{aligned} \sum_{t=1}^T \left(c_t(x_t^A, u_t^A) - c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \right) \\ \leq \frac{\alpha T \eta L}{(1-\alpha)^2} \left(\alpha L_u + \frac{L_x}{1-\alpha} \right) = \frac{\alpha T \eta L^2}{(1-\alpha)^2}, \end{aligned}$$

where we used $L = \alpha L_u + \frac{L_x}{1-\alpha}$ in the final step. $\square$

Proof of Theorem 1. Starting from (5) we have the following decomposition of the regret:

$$\begin{aligned} \text{Regret}(T) &= \sum_{t=1}^T c_t(x_t^A, u_t^A) \\ &\quad - \min_{M \in \Delta_H} \sum_{t=1}^T c_t(x_t^{\pi(M)}, u_t^{\pi(M)}) \\ &= \sum_{t=1}^T c_t(x_t^A, u_t^A) - \sum_{t=1}^T c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ &\quad + \sum_{t=1}^T c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ &\quad - \min_{M \in \Delta_H} \sum_{t=1}^T c_t(x_t^{\pi(M)}, u_t^{\pi(M)}). \end{aligned}$$

We call the first sum the stability term, and the second sum the OMD term. We will bound each term separately.

We bound the OMD term by unnormalized-negentropy OMD on $\Delta_H \subset \mathbb{R}^H$. Let $\Psi(M) = \sum_{i=1}^H (M^{(i)} \log M^{(i)} - M^{(i)})$ and let D_Ψ be its Bregman divergence as in Lemma 6. Under the initialization (13), for any benchmark $M^* \in \Delta_H$ with $W^* \triangleq \sum_{i=1}^H M^{*(i)} \leq 1$,

$$\begin{aligned} D_\Psi(M^* \| M_1) &= \sum_{i=1}^H \left(M^{*(i)} \log((H+1)M^{*(i)}) - M^{*(i)} + \frac{1}{H+1} \right) \\ &= \sum_{i=1}^H M^{*(i)} \log M^{*(i)} + W^* \log(H+1) - W^* + \frac{H}{H+1}. \end{aligned}$$

Since each $M^{*(i)} \in [0, 1]$, one has $M^{*(i)} \log M^{*(i)} \leq 0$, and therefore

$$D_\Psi(M^* \| M_1) \leq W^* \log(H+1) - W^* + \frac{H}{H+1}.$$

The right-hand side is at most $\log(H+1)$ for every $W^* \in [0, 1]$: equivalently,

$$(1 - W^*) \log(H+1) + W^* - \frac{H}{H+1} \geq 0,$$

which holds because the left-hand side is affine in W^* and nonnegative at the endpoints $W^* = 0$ and $W^* = 1$. Hence $D_\Psi(M^* \| M_1) \leq \log(H+1)$. The surrogate costs ℓ_t from (17) are convex and L -Lipschitz by Lemma 5, so the standard

OMD regret bound with Bregman divergence D_Ψ on Δ_H [19] yields

$$\sum_{t=1}^T \ell_t(M_t) - \min_{M \in \Delta_H} \sum_{t=1}^T \ell_t(M) \leq \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2}.$$

Using (17), we substitute $\ell_t(M) = c_t(x_t^{\pi(M)}, u_t^{\pi(M)})$, so that

$$\begin{aligned} \sum_{t=1}^T \ell_t(M_t) &= \sum_{t=1}^T c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}), \\ \min_{M \in \Delta_H} \sum_{t=1}^T \ell_t(M) &= \min_{M \in \Delta_H} \sum_{t=1}^T c_t(x_t^{\pi(M)}, u_t^{\pi(M)}). \end{aligned}$$

Thus this term becomes

$$\begin{aligned} &\sum_{t=1}^T c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ &- \min_{M \in \Delta_H} \sum_{t=1}^T c_t(x_t^{\pi(M)}, u_t^{\pi(M)}) \\ &\leq \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2}. \end{aligned}$$

From Lemma 13, the stability term satisfies

$$\begin{aligned} &\sum_{t=1}^T c_t(x_t^A, u_t^A) - \sum_{t=1}^T c_t(x_t^{\pi(M_t)}, u_t^{\pi(M_t)}) \\ &\leq \frac{\alpha \eta L^2 T}{(1-\alpha)^2}. \end{aligned}$$

Adding the OMD term and the stability term yields the combined bound

$$\text{Regret}(T) \leq \frac{\log(H+1)}{\eta} + \frac{\eta L^2 T}{2} + \frac{\alpha \eta L^2 T}{(1-\alpha)^2}.$$

Rearranging by factoring out the η term gives

$$\text{Regret}(T) \leq \frac{\log(H+1)}{\eta} + \eta L^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right).$$

To optimize over $\eta > 0$, we apply the inequality $A/\eta + B\eta \leq 2\sqrt{AB}$ with $A = \log(H+1)$ and $B = L^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right)$. This yields the optimal step size

$$\eta^* = \sqrt{\frac{\log(H+1)}{L^2 T \left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right)}}$$

and the resulting regret bound

$$\text{Regret}(T) \leq 2L \sqrt{\left(\frac{1}{2} + \frac{\alpha}{(1-\alpha)^2} \right) T \log(H+1)}.$$

□

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